

1. Extend  $\deg(0) = -\infty$ .

$$\begin{aligned} f &= qg + r \\ f(t) &= f_d t^d + \cdots, f_d \neq 0 \\ q(t) &= a_n t^n + \cdots, a_n \neq 0 \\ g(t) &= b_m t^m + \cdots, b_m \neq 0 \end{aligned}$$

For existence:

$$\begin{aligned} P(n): \text{ for every } f \in F[t] \text{ with } \deg(f) = n, n \in \mathbb{N}, g \neq 0, \exists q, r: f &= qg + r, \deg(r) < \deg(g) \\ f = 0: 0 &= 0 \cdot g + 0 \\ \deg(0) &= -\infty < \deg(g) \end{aligned}$$

- base

$$\begin{aligned} P(n) \text{ is true for } n < \deg(g) \\ q = 0, r = f, f &= qg + r, \deg(r) = \deg(f) = n < \deg(g) \end{aligned}$$

- induction step, assume for every  $n \leq N$ :  $P(n)$  is true, prove it for  $N + 1$ .

$$\begin{aligned} g_m \in F, F \text{ is a field, } g_m \neq 0 &\implies \exists g_m^{-1} \\ \text{let } a &= g_m^{-1} f_{N+1} \\ \text{let } f' = f - ag_m t^{N+1-m} &\implies \deg(f') < \deg(f) \implies f' = q'g + r \implies f = (q' + at^{N+1-m})g + r \end{aligned}$$

which concludes the induction.

For uniqueness assume that  $\exists q', r': q \neq q'$  or  $r \neq r', f = q'g + r'$ .

$$\begin{aligned} f &= qg + r, f = q'g + r' \\ qg + r - (q'g + r') &= 0 \\ (q - q')g + (r - r') &= 0 \\ \deg(r) < \deg(g), \deg(r') < \deg(g) &\implies \deg(r - r') < \deg(g) \implies \deg(r' - r) < \deg(g) \end{aligned}$$

Consider two cases:

$$q = q' \implies r = r'$$

contradiction with initial assumption

$$\begin{aligned} q \neq q' &\implies \deg(q - q') \geq 0 \implies \deg((q - q')g) \geq \deg(g) \\ (q - q')g &= r' - r \implies \deg((q - q')g) = \deg(r' - r) \end{aligned}$$

which is a contradiction since  $LHS \geq \deg(g), RHS < \deg(g)$ , therefore  $q' = q, r' = r$ .